

todo Co-Formation Games

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Abstract—This report proposes a novel game theoretical model combining rational network formation and opinion dynamics. After introducing the fields of both Network Formation Games (NFGs) and Social Influence Models (SIMs), I compose a model allowing for co-evolution of opinions and networks and containing both NFGs and SIMs as special cases. Further sections consider basic dynamics, optimal states, and research directions of these Co-Formation Games (CFGs), next to showcasing a CFG simulations module. Overall, the model broadens the perspective of NFGs and SIMs, offering a promising basis for future research projects.

Index Terms—Collective dynamics, Game theory, Network formation games, Network theory (graphs)

I. INTRODUCTION

A. Paper overview

todo

II. METHODS

A. Definitions & Equations

TABLE I
LIST OF VARIABLES USED IN THIS DOCUMENT

Variable	Symbol	Domain
Number of nodes	n	$\mathbb{N}$
Node of concern	i	$\{1, \dots, n\}$
Set of i 's neighbours	K_i	$\mathbb{N}^{ K_i }$
Network graph ¹	G	$\{0, 1\}^{n \times n}$
Opinion	x_i	$\mathbb{R}$
Action profile	a_i	$\{0, 1\}^n$
Costs per edge	α_c	$[0, \infty[$
Pairwise distance	d_{ij}	$\mathbb{N}$
Network valuation	v	$] - \infty, 0]$
Network expenses	f^c	$] - \infty, 0[$
Cost function	c	$] - \infty, 0[$
Communication rate	r_c	$[0, \infty[$
Formation rate	r_f	$\{0, 1\}$
Convergence threshold	ε	$[0, \alpha_c]$

where a norm $\|\cdot\|$ is the L0 norm, returning the count of non-zero elements.

¹The graph is represented as an adjacency matrix.

a) Equations:

- Cost function

$$c_i = v_i(a) + f^c(a) \quad (1)$$

- Valuation function

$$v_i(a) = - \sum_{j=1, \dots, n} d_{ij} \quad (2)$$

- Network expenses:

$$f_i^c(a) = -\alpha_c \|a_i\| - \frac{1}{\|K_i\|} \sum_{j \in K_i} |x_i - x_j| \quad (3)$$

b) Algorithm:

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1 Initialise  $t = 1, c$ 
2 while ( $t < t_{\max} \wedge \varepsilon < \Delta c$ )
3   if  $\text{mod}(t, r_c) = 0$  then
4     select random node  $i$ 
5      $x_i$  arrow.l get opinion update  $x_i^u$ 
6   end if
7   if  $r_f = 1$  then
8     select random node  $i$ 
9      $a_i$  arrow.l get action update  $a_i^u$ 
10  end if
11   $t += 1$ 
12   $c(t)$  arrow.l get new costs  $c_t$ 
13   $\Delta c = c_{t-1} - c(t)$ 
14 end while

```

Algorithm 1. Dynamics of our synchronous move game

III. OLD STUFF FROM THE TEMPLATE



Fig. 1. A circle representing the Sun.

In Fig. 1 you can see a common representation of the Sun, which is a star that is located at the center of the solar system.

In Table II, you see the planets of the solar system and their average distance from the Sun. The distances were calculated with (1) that we presented in Section II.

TABLE II
THE PLANETS OF THE SOLAR SYSTEM AND THEIR AVERAGE DISTANCE FROM
THE SUN

Planet	Distance (million km)
Mercury	57.9
Venus	108.2
Earth	149.6
Mars	227.9
Jupiter	778.6
Saturn	1,433.5
Uranus	2,872.5
Neptune	4,495.1

REFERENCES